# What are the Protocols and Resources for Sport-Related Concussion Among Top National Collegiate Athletic Association Football Programs? A Cross-Sectional Survey of A5 Schools

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**Abstract**

**Objective:** This study summarizes findings from a cross-sectional survey conducted among National Collegiate Athletic Association (NCAA) Division 1 football programs, focusing on sport-related concussion (SRC) protocols for the 2018 season. **Design:** Cross-sectional survey study. **Setting:** 65 football programs within the Autonomy Five (A5) NCAA conferences. **Participants:** Athletic trainers and team physicians who attended a football safety meeting at the NCAA offices June 17 to 18, 2019, representing their respective institutions. **Intervention:** Electronic surveys were distributed on June 14, 2019, before the football safety meeting. **Main Outcome Measures:** Results for 16 unique questions involving SRC protocols and resources were summarized and evaluated. **Results:** The survey garnered responses from 46 of 65 programs (response rate = 71%). For *baseline testing*, 98% measured baseline postural stability and balance, 87% used baseline neurocognitive testing, while only 61% assessed baseline vestibular and/or ocular function. Regarding *concussion prevention*, 51% did not recommend additional measures, while 4% and 24% recommended cervical compression collars and omega-3 supplementation, respectively. In *postconcussion treatment*, 26% initiated aerobic exercise 1 day postconcussion if symptoms were stable, 24% waited at least 48 hours, 4% waited for the athlete to return to baseline, 11% waited until the athlete became asymptomatic, and 35% determined procedures on a case-by-case basis. **Conclusions:** Most institutions assessed postural stability/balance and neurocognitive functioning at baseline and introduced light aerobic exercise within 48 h postconcussion. There was variation in baseline assessment methods and concussion prevention recommendations. These survey findings deepen our understanding of diverse SRC protocols in NCAA football programs.

**Key Words:** sport-related concussion, football, universities, survey

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## INTRODUCTION

American football continues to have one of the highest incidence[1] and prevalence[2] of sport-related concussion (SRC) among all National Collegiate Athletic Association (NCAA) sports. The findings[2] from the NCAA-Department of Defense (Department of Defense) Concussion Assessment, Research, and Education (CARE) Consortium showed that across a total of 1751 SRCs in 22 different collegiate sports between 2014 and 2018, 35% occurred in football. As such, efforts have been made to improve SRC management, policies, and awareness in collegiate football over the years.[3–6] In a recent study[3] comparing football data from the NCAA Concussion Study (1999-2001) and the NCAA-DoD CARE Consortium (2014-2017), CARE football athletes showed a longer symptom duration, longer symptom-free waiting period, and longer return-to-play compared with NCAA Concussion Study football athletes. In addition, while the average time between first and repeat SRC was 6 days in the NCAA Study, it was 60 days in CARE.[3] Despite major improvements in the management of SRC in collegiate football,[5] additional information on resources and practices regarding SRC treatment and prevention remains to be elucidated.[4,6]

In 2019, head football athletic trainers, primary care team physicians, and orthopedic team physicians from the football



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programs of the NCAA Division 1 Autonomy Five Conferences (A5) convened for a football safety meeting jointly sponsored by the National Football League (NFL) and the NCAA.[7–9] The A5 conferences include the Southeastern, Atlantic Coast, Big Ten, Big 12, and Pacific 12 Conferences who were each granted autonomy by the NCAA Division 1 Board of Directors to create their own legislation.[10] The objective of the meeting was to facilitate exchange of ideas between sports medicine experts who directly participate in the care of football athletes at the A5 level.[9] To the best of our knowledge, this meeting was the first one of its kind. Before the meeting, each participating institution was asked to complete a survey that assessed the current healthcare protocols and available resources in the following 6 main areas: (1) SRC, (2) emergency action planning, (3) mental health, (4) orthopedic injuries, (5) preparticipation evaluation, and (k6) infrastructure.[7–9] The results of this topical survey provided information about institutional protocols that may serve as valuable normative data and benchmarks for future research efforts.

Using the cross-sectional survey data of American football programs of the NCAA Division 1 Autonomy Five Conferences, we sought to summarize the protocols and available resources specific to SRC in the 2018 season. Areas of SRC protocols and resources were divided into the following: (1) assessment, (2) prevention, and (3) management.

## METHODS

### Study Design

A cross-sectional survey was undertaken of all NCAA A5 football programs attending a football safety meeting jointly sponsored by the NFL and the NCAA, convened June 17 to 18, 2019, in Indianapolis, Indiana.[7–9] The meeting agenda encompassed 6 key topics: (1) SRC, (2) emergency action planning, (3) mental health, (4) orthopedic injuries, (5) preparticipation evaluation, and (6) infrastructure.[7–9] Discussions aimed to solidify best practices and help attendees understand normative practices of A5 programs, and the results pertaining to SRC were the focus of this article.

### Participants

Participants included head football athletic trainers, primary care team physicians, and orthopedic team physicians from the 65 football programs of the NCAA A5 conferences.[8] The overall response rate among the 65 institutions surveyed was 71% (46/65 programs), with responses to each question ranging from 45% (29/65) to 71% (46/65). Demographic data including the participants' age, sex, and role were not available for this study.

### Survey

Three days before the commencement of the meeting, each institution's participants were sent an electronic survey composed of 143 unique questions across the 6 aforementioned topics. Only 1 invitation to fill out the survey was provided during the study period, without any reminder attempts. Participants were instructed to discuss the questions with other participants from their respective institution before responding, and only 1 aggregate response from each institution was allowed. There were 16 questions which pertained to (1) baseline assessment, (2) prevention, and (3) management of SRC (see **Appendices, Supplemental Digital Content 1-3**, http://links.lww.com/JSM/A421, http://links.lww.com/JSM/A422, http://links.lww.com/JSM/A423) in their football athletes during a one-year period (from January 2018 through the end of 2018 competitive season, including bowl games, which may have occurred as late as January 2019). Descriptive statistics were performed on these survey results.

## RESULTS

### Concussion Baseline Assessment

Regarding baseline testing, nearly all schools (n = 45/46, 98%) completed some form of assessment. Baseline evaluation of postural stability/balance was conducted in all football programs that performed testing (n = 45/46, 98%). Regarding modality, when asked which specific assessment was used to measure postural stability/balance (with multiple answers allowed), the Balance Error Scoring System (BESS) was the most commonly used (n = 19/55, 35%), followed by device-based measurement (Sway Balance, NeuroCom, force plate; n = 16/55, 29%), Modified BESS (n = 15/55, 27%), Timed Tandem Gait (n = 4/55, 7%), and other (n = 1/55, 2%). Across institutions, 87% (n = 40/46) of programs incorporated computerized neurocognitive testing into their baseline evaluations while 13% (n = 6/46) did not. The most prevalent modality used (with multiple answers allowed) was ImPACT testing (n = 27/42, 64%), followed by C3 Logix (n = 7/42, 17%), Axon Sports/CogState (n = 3/42, 7%), other (n = 3/42, 7%), and CNS Vital Signs (n = 2/42, 5%). Of all programs surveyed, 61% (n = 28/46) completed baseline vestibular and/or ocular function while 39% (n = 18/46) did not. Regarding modality (with multiple answers allowed), eye-tracking devices (SyncThink Eye-Sync/Oculogica EyeBOX) (n = 12/36, 33%) were most common, followed by Vestibular Ocular Motor Screen (VOMS) (n = 11/36, 31%), King-Devick Test (n = 10/36, 28%), other (n = 2/36, 6%), and convergence measurement alone without other VOMS components (n = 1/36, 3%). When asked whether they used other assessment modalities, 1 institution reported using electroencephalogram (1/45, 2%), while no institutions used speech recognition or auditory testing as part of their baseline assessments. Altogether, 30% (n = 14/46) of institutions selected assessments that were outside (ie, in addition to) their institution's NCAA-approved concussion management plan. The frequency of baseline testing for concussion assessment varied across institutions: every year (n = 8/46, 17%), first year on campus only (n = 18/46, 39%), every other year (n = 6/46, 13%), any of the previous options plus 6 months or other interval after a concussion (n = 12/46, 26%), and other (n = 2/46, 4%). Concussion baseline assessment results are shown in Figure 1.

### Concussion Prevention

Regarding preventative measures for concussion (multiple answers allowed), nearly half of schools (25/51, 49%) recommended preventative measures for concussion to



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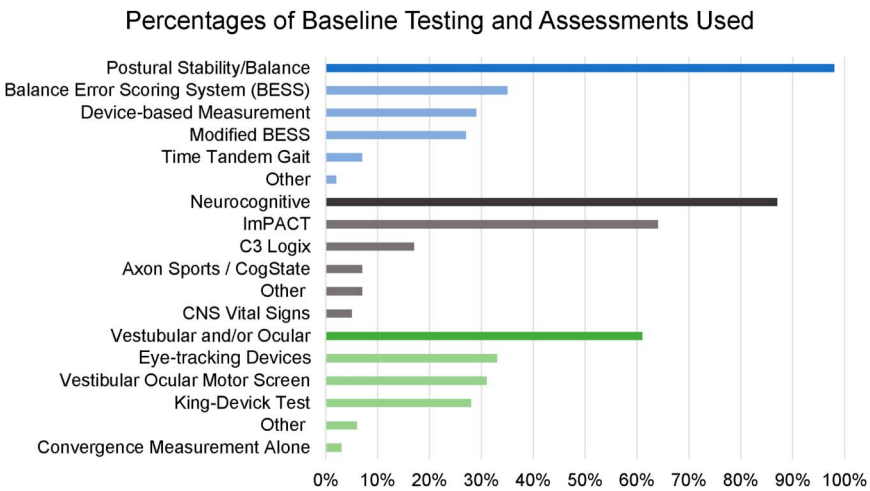


**Figure 1.** Bar graph showing the percentages of each postural stability/balance, neurocognitive, and vestibular and/or ocular baseline testing and the assessments used within each.

athletes. Of these programs, 24% (n = 12/51) recommended that their football athletes try prophylactic use of omega-3 supplements, 22% (n = 11/51) recommended external helmet padding (Guardian Caps), and 4% (n = 2/51) recommended cervical compression collar (Q-collar). To gain information about the potential for athletes to encounter differences in helmet fit or helmet type throughout the year, 59% (n = 27/46) of institutions reported using different helmet types for practice versus games. Of this group, 93% (25/27) used helmets of the same make and model between practice and games, while 7% (n = 2/27) did not. Regarding helmet styles, 74% (n = 34/46) of the teams used more than 1 helmet style for games in 2018 (eg, different color, sheen, and lettering), while 26% (n = 12/46) did not. Of the teams that used different helmet styles, 97% (n = 33/34) used helmets that were all the same make and model. Concussion prevention data are shown in Figure 2.

### *Concussion Management*

When asked about the timing of starting light aerobic exercise after a concussion diagnosis, different approaches were reported. Among the programs, 26% (n = 12/46) reported that they would initiate light aerobic exercise on the day after the concussion if the athlete's symptoms were stable and they could be monitored for any exacerbation of symptoms and 24% (n = 11/46) reported waiting for 48 hours after concussion and until symptoms are stable. A minority of schools indicated that they would begin aerobic exercise once the athlete becomes asymptomatic (11%, n = 5/46) and 4% (n = 2) reported waiting until the athlete returns to their baseline to initiate aerobic exercise. Finally, approximately one-third of schools reported aerobic exercise initiation was determined on a case-by-case basis (35%; n = 16/46). Concussion management data are shown in Figure 3.

## DISCUSSION

Using cross-sectional survey data from 46 of 65 American football programs within the NCAA Division 1 Autonomy Five Conferences, our study describes protocols and resources commonly used for the baseline assessment, prevention, and management of SRC during the 2018 season. The study found that most institutions administered baseline postural stability/balance and neurocognitive tests, although baseline vestibular and/or ocular function testing was less commonly conducted. Furthermore, approximately half of programs recommended preventative measures, such as the cervical compression collar and omega-3 fatty acid supplementation, to athletes as part of

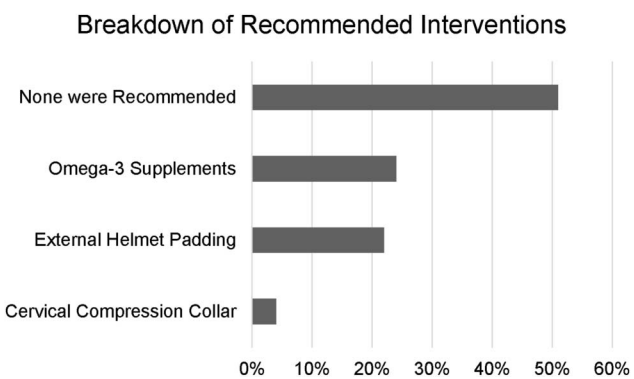


**Figure 2.** Bar graph displaying the percentages of recommended interventions for concussion prevention.

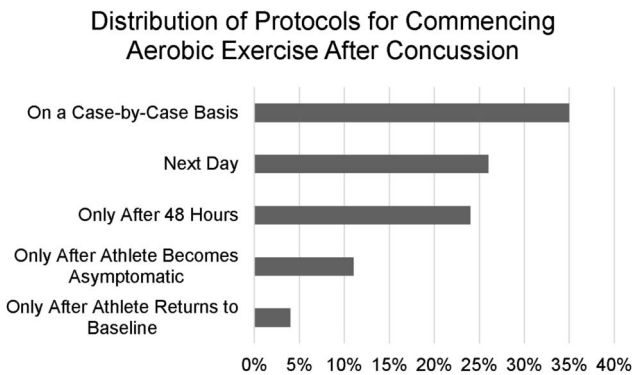


**Figure 3.** Bar chart illustrating the distribution of implemented protocols for initiating light aerobic exercise after a concussion, presented as percentages.



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SRC prevention strategies. Notably, there existed significant heterogeneity in protocols regarding the initiation of aerobic exercise postconcussion, with only half of programs starting aerobic activity within 48 hours, and more than a third of programs reporting that return to physical activity was determined on a case-by-case basis.

### Baseline Assessment

Baseline testing is a common facet within SRC management at the A5 level, providing healthcare professionals with the means to quantify the extent of postconcussion changes in an individual and thus tailor treatment strategies accordingly.[11] In addition, a return of cognitive function, physical capabilities (eg, balance), and self-reported symptoms to baseline levels serves as a guide for an athlete's return-to-play timeline.[12] During the 2018 to 2019 season, the NCAA mandated that programs establish preparticipation baseline testing procedures in accordance with the Inter-Association Consensus: Diagnosis and Management of SRC Best Practices[13] (bylaw amendment 3.2.4.17.1).[14] The Inter-Association Consensus, a collaborative effort between the NCAA Sports Science Institute and various sports medicine organizations, was formulated during the Safety in College Football Summit in February 2016.[13,15] Regarding baseline assessments, the Inter-Association Consensus[13] recommended, at minimum, the utilization of neurocognitive and postural stability/balance testing, which was common among the surveyed programs. However, only 61% implemented vestibular and/or oculomotor baseline assessments. Despite previously established literature recognizing the role of vestibular and oculomotor symptoms in SRC, and the reliability of various assessment methods, the Inter-Association Consensus did not specifically recommend baseline testing for these functions.[13,16–21] Establishing an athlete's vestibulo-ocular status before the season's commencement assumes particular importance, as highlighted in later work by Moran et al[22] who noted that nearly 20% of nonconcussed youth athletes exhibited nonzero scores on the VOMS scale. Furthermore, Iverson et al[23] have since found that approximately 29.3% of youth athletes surpassed clinical thresholds in aspects of the VOMS, underscoring the risk of misattributing longstanding vestibulo-ocular issues, if newly observed postconcussion, to the concussion itself. For programs that conducted baseline vestibular/oculomotor testing, tools such as the VOMS, eye-tracking devices, and the King-Devick test were widely used. It is interesting that 30% of programs reported using baseline tests/modalities with their football program which were outside their institution's concussion management plan. Whether this indicates additional focus on SRC for their football athletes or differences in resource allocation compared with other sports cannot be ascertained by this study.

### Prevention

Prevention strategies were used in half of schools, with prophylactic omega-3 supplements being the most popular recommendation, followed by external helmet padding and cervical compression collars. Notably, neither the NCAA Inter-Association Consensus nor the 2017 concussion in sport group (CISG) consensus statement makes mention of these interventions.[13,20] Cervical compression collars were developed to counteract the movement of brain tissue and fluid within the cranial cavity due to dynamic forces by applying gentle pressure to the jugular veins thereby increasing intracranial blood volume and restricting excessive movement.[24,25] The objective of reducing brain movement during/after head impact is to uphold the integrity of white matter, which is important in facilitating proper maturation crucial to cognitive, emotional, behavioral, and motor development.[26] Preceding the 2018 to 2019 season, various preclinical laboratory and pilot clinical studies involving cervical compression collars had been conducted.[27–29] A study by Myer et al[24] observed significant reductions in the diffusion coefficient, a marker of white matter integrity, within high-school football athletes who did not use cervical compression collars, whereas no such reductions were evident among athletes who used the collars. Importantly, the clinical implications of the white matter findings are unknown because no correlation with symptoms or neurocognitive tests were performed. Despite suggestions that cervical compression collars can mitigate white matter injury stemming from subconcussive head impacts,[24,25] empirical evidence, both before the 2018 to 2019 season and at present, does not substantiate a concurrent reduction in the risk of SRC or the severity of head impacts.[24,30]

Regarding the potential neuroprotective effects of omega-3 fatty acid supplementation, these benefits have been proposed and encompass stabilization of homeostasis and the promotion of anti-inflammatory responses after traumatic brain injuries.[31] Before the 2018 to 2019 season, preclinical rat studies indicated that omega-3 fatty acid supplementation reduced axonal injury when given either prophylactically to[32] or after[33] induction of impact acceleration traumatic brain injury, leading to the hypothesis that supplementation might mitigate concussions.[34] In addition, omega-3 supplementation was shown to effectively dampen elevations in serum neurofilament light, a biomarker indicative of neuroaxonal injury subsequent to traumatic brain injury and subconcussive head impacts, in football athletes.[35] Nonetheless, more recent research by Raikes et al[36] found no substantial changes in white matter integrity as assessed through diffusion tensor imaging within a cohort of NCAA American football athletes. Consequently, while standard dosing of omega-3 fatty acids is generally considered safe, the body of evidence insufficient to endorse the prophylactic utilization of omega-3 fatty acids for SRC prevention.[31,37] Despite their well-intentioned nature, these proposed interventions for preventing concussions may inadvertently embolden athletes to undertake higher-risk actions under the guise of enhanced protection, underscoring the importance of educating coaches, staff, and athletes on the limitations associated with these interventions. Although omega-3 fatty acids are fairly benign, and at worst add helpful nutrients to the body, data regarding the effectiveness of their ability to reduce concussions are absent.

The current literature on the efficacy of external padding, such as the Guardian Cap, in preventing concussions remain highly limited. Leading up to the 2018 to 2019 season, very few studies had examined the effects of external padding through drop tests and impact kinematics and revealed limited support for the use of external padding to mitigate concussion risk or severity.[38,39] Subsequent studies have since been conducted, including 1 during the 2022 preseason where the NFL mandated the use of the Guardian Cap for all NFL players up until the second preseason game.[40] Although data from this study have yet to be published, the NFL stated



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a reduction of more than 50% in concussions compared with the averages of the preceding 3 years.[40] The Virginia Tech Helmet Lab[41] found that helmet shell add-ons decrease both linear and rotational acceleration in the laboratory setting, potentially reducing the risk of SRC. However, the study did not report detailed statistical analyses and highlighted that the efficacy of the external add-ons depended on the base helmet model.[41] Bailey et al[42] also measured impact severity differences with and without external padding across multiple helmet designs in the laboratory setting. They reported that while external padding reduced head impact severity for most helmet designs, helmets with better baseline performance were minimally improved by add-ons. In a separate study, Cecchi et al[43] used instrumented mouthguards to evaluate the performance of external helmet add-ons in 5 linebackers over 2 seasons, 1 played with bare helmets and 1 played with helmet add-ons, and did not find that wearing a helmet add-on significantly reduced on-field head impact magnitude compared with bare helmet controls. Similarly, a study with NCAA football players showed no significant differences in head movement with the use of Guardian Caps as compared with their performance without Guardian Caps.[44] Collectively, the outcomes of these investigations suggest a need for further on-field assessments involving larger sample sizes to gain a better understanding of the role external padding plays in concussion risk.

A concern has emerged regarding the potential risks associated with the practice of switching between different helmets throughout a season either when using one helmet for practices and another for games or when using helmets with distinct appearances for specific games.[45] Our findings demonstrate that it is common to use different helmets for practice and games despite most institutions using the same size, make, and model for each player. Similarly, the trend of using different game helmets throughout the season is also prevalent. Even in this scenario, athletes are typically provided with helmets of the same make, model, and size.

### *Active Rehabilitation*

More than one-third of institutions have reported adopting a personalized approach to initiating light aerobic exercise for athletes, making decisions on a case-by-case basis. A robust body of research underpins the significance of early active rehabilitation in accelerating recovery postconcussion and mitigating the chances of persistent symptoms.[46,47] The 2017 CISG consensus statement advocated for a brief period of rest (24-48 hours) before gradually progressing through subthreshold activity.[13,20] In addition, the NCAA Inter-Association Consensus emphasized the importance of using an individualized approach to return-to-play protocols, potentially contributing to the diversity of responses recorded in the survey.[13] Preceding the 2018 to 2019 season, the research on early active rehabilitation was novel and continues to be an area of growing interest. Current research suggests that restricting physical activity could lead to physical deconditioning and secondary symptoms in athletes.[48,49] Furthermore, physical inactivity is believed to extend recovery timelines for various conditions often comorbid with concussions, such as vestibular disorders, depression, and fatigue. However, it is important to tread cautiously: although an immediate return to competitive sports and intense exercise after concussion is discouraged, engaging in physical activity with controlled intensity and duration, and a focus on minimizing symptom exacerbation, can be safely initiated as early as 1 day postconcussion.[50] More recent works reaffirm literature available before the 2018 to 2019 season, indicating that after the initial days postconcussion, it is both safe and efficacious to commence aerobic exercise at a subsymptom heart rate threshold, even in symptomatic athletes.[12,51] Given the potential for prolonged recovery and enduring symptoms to disrupt an athlete's engagement in academic, social, and athletic pursuits, it is important to implement concussion management protocols that encompass early physical activity and aerobic exercises.[52]

### *Future Directions in Concussion Care*

During the 2018 to 2019 season, the NCAA provided programs with a broad concussion safety framework, granting them autonomy in protocol design. The current survey revealed that there were substantial variations in baseline measurements, return-to-activity schedules, and overall concussion care and management across programs. Although individualization of concussion protocols based on the specific needs and resources of their athletes and staff is important, the lack of standardization may result in disparities in the quality of care, hinder cross-program data comparison, or create lack of agreement among coaches and athletes. Moreover, the diversity in concussion care may render some programs unaware of the latest evidence-based best practices, potentially leading to suboptimal care for athletes. Despite the NCAA's directive for programs to align their concussion safety protocols with the 2016 Inter-Association Consensus, it is advisable for programs to regularly analyze the literature for new updates to ensure the ongoing relevance and effectiveness of concussion management protocols. Moving forward, it may be advantageous for the NCAA or individual A5 conferences to consider standardizing concussion care on a larger scale and facilitating communication on novel research findings between programs. A collaborative effort could contribute to a more cohesive and evidence-driven approach to concussion management across collegiate sport programs.

Survey responses regarding potential concussion prevention strategies exhibited significant variability, presenting a unique scenario not specifically addressed by the NCAA bylaws, Inter-Association Consensus, or the 2017 CISG consensus statement.[13,14,20] Implementing prevention methods requires careful consideration and evaluation. Programs advocating for prevention strategies should thoroughly assess the literature and prioritize evidence-informed decision-making. A comprehensive risk-benefit analysis and consultation with experts are essential to evaluate the impact on the health and safety of athletes.

The insights gleaned from this survey offer valuable directions to enhance both clinical care and research concerning collegiate athletes. The following key points encapsulate potential avenues for advancement:

1. A broader adoption of vestibular and ocular assessments at baseline within NCAA programs is advisable. Expansion of baseline testing could provide more comprehensive insights into athletes' postconcussion conditions.
2. Regarding prevention, with newer data regarding the Guardian Cap at the professional level, this should be



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considered by collegiate teams and medical staff while investigations on its effects continue. Future research endeavors should delve deeper into the use of cervical compression collars and omega-3 supplementation in SRC prevention and recovery and explore the potential behavioral implications of using these interventions during athletic participation.

3. As a practical guideline, programs are recommended to consider implementation of subthreshold physical activity within the first day and controlled aerobic exercise within 2 days postconcussion to facilitate recovery and protect against persistent symptoms. These are in accordance with the most recent CISG guidelines.[12]
4. It is worth noting that this study can serve as a reference point for assessing the evolution of protocols and safety measures adopted by the NCAA subsequent to 2018.

In essence, these directions offer actionable steps that can collectively contribute to the advancement of concussion management and further the well-being of collegiate athletes.

### *Limitations*

Although this study has provided valuable insights, it is essential to acknowledge its inherent limitations. Most importantly, the survey data were collected for the 2018 season and do not necessarily reflect the protocols used at present. Second, the scope of our analysis encompassed survey responses from 46 of 65 NCAA A5 institutions. It is important to acknowledge that the response rate for each question exhibited variability, ranging from as low as 45% to as high as 71% for primary questions. The variability in response rates implies that the data presented might not offer a fully representative view of all NCAA A5 institutions. Third, a potential source of bias lies in the unavailability of demographic data for respondents who participated in the survey. The absence of demographic information can introduce an element of bias into the findings and interpretations. Fourth, it is crucial to recognize that the results presented in this study are solely descriptive in nature. The data available do not support drawing definitive conclusions about the relationships that may exist between institutions and their respective responses. Finally, the survey used multiselect questions, which may introduce complexity when attempting to discern variations in resource utilization across institutions. These limitations underscore the necessity for cautious interpretation and the recognition that the findings may not comprehensively capture the intricacies present within the broader landscape of NCAA institutions.

## CONCLUSIONS

Through an in-depth analysis of survey data from 46 of 65 NCAA A5 football programs during the 2018 season, we explored variations in SRC evaluation and management. Across programs, the survey results showed substantial variations in baseline measurements, return-to-activity schedules, and overall concussion care. Although many programs used baseline postural stability and neurocognitive testing, assessing baseline vestibular and ocular assessments were less common. Some programs recommended cervical compression collars, omega-3 fatty acid supplementation, and external padding for concussion prevention, warranting future research on their impacts. In addition, around one-third of surveyed programs determined postconcussion initiation of aerobic exercise case-by-case. Notably, existing research supports early physical activity initiation during recovery. Our study underscores gaps and avenues for further research, contributing to improved concussion care strategies in collegiate athletics.

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